1	2	3	4	5	6	Total
/20	/20	/20	/20	/20	/20	/100

Name: ______________________________ April 20, 2012 Friday

Department/School:______________________ SID:____________________________

University Physics

Midterm Examination

Kuang Yaming Honors School, Nanjing University

Select five out of following six problems.

1. (20pts) A sphere of mass m moves under uniform gravity and in a medium that exerts a resistance force that is proportional to the square of its speed. When the sphere falls vertically in the medium, its terminal velocity is v_0. The sphere is now projected vertically upward with an initial speed u.

(a) Write the equation of motion of the sphere.

(b) Find the maximum height the sphere can reach in terms of g, v_0 and u.

(c) Find the time t_m the sphere reaches its maximum height.

2. (20 pts.) A particle of mass m moves in a one-dimensional region $x > 0$. The potential energy of the particle is

$$U(x) = \frac{1}{2} m\omega^2 \left(x^2 + \frac{a^4}{x^2} \right)$$

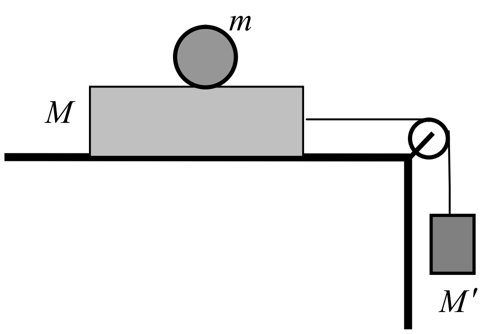
where ω and a are positive constants.

(a) Find the force acted on the particle.

(b) Find the equilibrium position of the particle and identify it as stable or unstable.

(c) If the particle starts at the equilibrium position with velocity v, find the equation that determines the extreme positions of the particle.

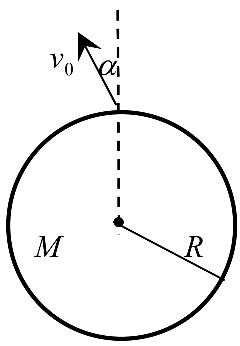
3. (20pts.) As shown in the accompanying figure, a uniform ball of mass m and radius R stands on a block of mass M that can slide without friction on a horizontal table. The block is connected to a hanging mass M' by a massless string passing over a massless and frictionless pulley. Assume the ball does not slip on the top of the block.



(a) What are the translational accelerations of the block and the center of mass of the ball?

(b) What is the rotational acceleration of the ball?

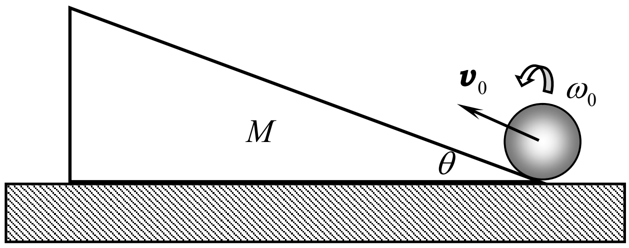
4. (20 pts.) A spacecraft of mass m is launched from the surface of a non-rotating spherical planet of mass M and radius R with an unknown speed v_0 at an angle of α to the local vertical. (Assume that the burn time of the rocket engine is very short and the spacecraft reaches speed v_0 immediately after the ignition.) It is found that the spacecraft travels in a bound orbit and it reaches a maximum distance of R_m from the center of the planet.



(a) Find the initial speed of spacecraft v_0 and the speed of the spacecraft when it reaches its maximum distance from the center of the planet.

(b) What is the minimum initial speed the spacecraft required so that its orbit is not bound?

5. (20 pts.)A cylinder of mass m and radius R rolls up a wedge of angle θ without slipping. The wedge is on a frictionless horizontal surface. The mass of the wedge is M . Initially the cylinder moves with velocity v_0 and the wedge is at rest on the horizontal surface.



(a) How high can the cylinder roll up the wedge?

(b) When the cylinder reaches the highest point on the wedge, what is the velocity of the wedge at that time?

6. (20pts) As shown in the figure, a uniform circular disk of mass M and radius R is suspended at a point P which is at distance d ($d \le R$) from the center of the disk so that it can oscillate freely in the vertical plane of the disk.

(a) What is the rotational inertia of the disk about the axis through P? What is the torque acted on the disk relative to the point P?

(b) Write the equation of the rotational motion of the disk.

(c) What is the period of small oscillations?

(d) Suppose the suspending point P is adjustable so that the distance d can be changed. Compute the distance d at which the period of the small oscillations has the smallest value. What is the smallest period?